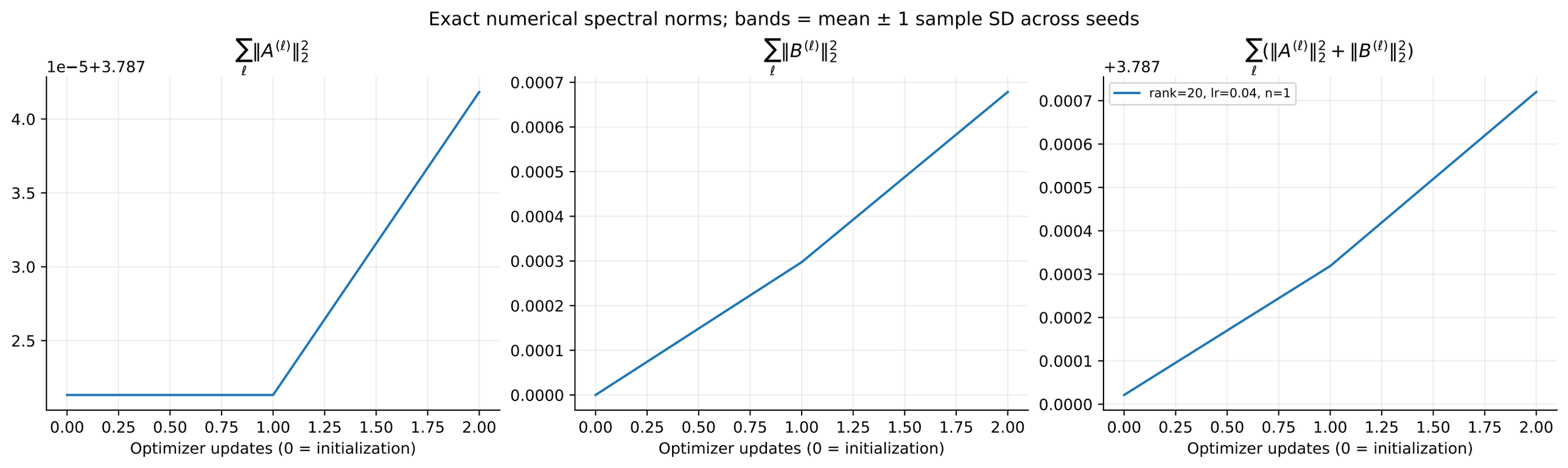
Exact numerical spectral norms; bands = mean ± 1 sample SD across seeds
$\sum_\ell \|A^{(\ell)}\|_2^2$
1e−5+3.787
4.0
3.5
3.0
2.5
$\sum_\ell \|B^{(\ell)}\|_2^2$
0.0007
0.0006
0.0005
0.0004
0.0003
0.0002
0.0001
0.0000
$\sum_\ell (\|A^{(\ell)}\|_2^2 + \|B^{(\ell)}\|_2^2)$
+3.787
rank=20, lr=0.04, n=1
0.00 0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00
Optimizer updates (0 = initialization)